UNIVERSITY OF SOUTHAMPTON AICE3001W1

SEMESTER 1 EXAMINATION 2026/27

MATHEMATICS OF MACHINE LEARNING (MOCK EXAMINATION)

DURATION 120 MINS (2 HOURS)

This paper contains FOUR questions.

Mock examination. Answer ALL parts of the question in section A (40 marks) and TWO questions from section B (30 marks each)

An outline marking scheme is shown in square brackets to the right of each question.

You ***must*** write your Student ID in the box and must not write your name anywhere on the paper.

Your answers should be written in the boxes provided on the question paper. The "Additional Space" provided on the facing or adjacent page to the question may be used for corrections or extra space. Clearly indicate which question the answer corresponds to.

Scrap paper can be used - this will be discarded without being looked at.

University approved calculators MAY be used.

Student ID:

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Question	Mark	*Arithmetic checked*	*Double Marked*
A1	/40		
B2	/30		
B3	/30		
B4	/30		
Total:			

Answers

A foreign language translation dictionary (paper version) is permitted provided it contains no notes, additions or annotations.

This examination does not contain colour figures.

15 page examination paper. This paper has no handouts.

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Section A

A1

(a) Using the convention that $(\partial g/\partial \mathbf{X})_{ij} = \partial g/\partial X_{ij}$, show that for a constant matrix $\mathbf{A}$

$$\frac{\partial \operatorname{tr}(\mathbf{A}\,\mathbf{X})}{\partial \mathbf{X}} = \mathbf{A}^{\mathsf{T}} \qquad \frac{\partial \operatorname{tr}(\mathbf{X}^{\mathsf{T}}\mathbf{A}\,\mathbf{X})}{\partial \mathbf{X}} = (\mathbf{A} + \mathbf{A}^{\mathsf{T}})\,\mathbf{X}.$$

What does the second result become when $\mathbf{A}$ is symmetric?

Writing the trace in components, $\operatorname{tr}(\mathbf{AX}) = \sum_{i,j} A_{ij}X_{ji}$, so differentiating with respect to X_{lm} picks out the single term with $j = l$, $i = m$, giving A_{ml}. That is the (l, m) element of $\mathbf{A}^{\mathsf{T}}$.
For the second, $\operatorname{tr}(\mathbf{X}^{\mathsf{T}}\mathbf{AX}) = \sum_{i,j,k} X_{ji}A_{jk}X_{ki}$. The index pair (l, m) can match either factor of $\mathbf{X}$:

$$\frac{\partial}{\partial X_{lm}} \sum_{i,j,k} X_{ji}A_{jk}X_{ki} = \sum_{k} A_{lk}X_{km} + \sum_{j} X_{jm}A_{jl} = (\mathbf{AX})_{lm} + (\mathbf{A}^{\mathsf{T}}\mathbf{X})_{lm}\,.$$

Hence the derivative is $(\mathbf{A} + \mathbf{A}^{\mathsf{T}})\,\mathbf{X}$, which reduces to $2\,\mathbf{A}\,\mathbf{X}$ when $\mathbf{A}$ is symmetric.

[5 marks]

(b) An inner product satisfies $\langle \boldsymbol{x}, \boldsymbol{x}\rangle \geq 0$ for all $\boldsymbol{x}$. By expanding the quadratic

$$q(t) = \langle \boldsymbol{x} + t\,\boldsymbol{y}, \boldsymbol{x} + t\,\boldsymbol{y}\rangle = A\,t^2 + 2\,B\,t + C$$

identify A, B and C, and by requiring that $q(t)$ never becomes negative deduce the Cauchy–Schwarz inequality

$$\langle \boldsymbol{x}, \boldsymbol{y}\rangle^2 \leq \langle \boldsymbol{x}, \boldsymbol{x}\rangle\,\langle \boldsymbol{y}, \boldsymbol{y}\rangle\,.$$

Expanding using bilinearity and symmetry of the inner product,

$$q(t) = \langle \boldsymbol{x}, \boldsymbol{x}\rangle + 2\,t\,\langle \boldsymbol{x}, \boldsymbol{y}\rangle + t^2\,\langle \boldsymbol{y}, \boldsymbol{y}\rangle\,,$$

so $A = \langle \boldsymbol{y}, \boldsymbol{y}\rangle$, $B = \langle \boldsymbol{x}, \boldsymbol{y}\rangle$ and $C = \langle \boldsymbol{x}, \boldsymbol{x}\rangle$.
Since $q(t) \geq 0$ for every real t, the quadratic cannot cross the axis, so it has no distinct real roots. From the standard solution $t = \left(-B \pm \sqrt{B^2 - A\,C}\right)/A$ this requires the discriminant to be non-positive,

$$B^2 - A\,C \leq 0 \qquad \Longrightarrow \qquad \langle \boldsymbol{x}, \boldsymbol{y}\rangle^2 \leq \langle \boldsymbol{x}, \boldsymbol{x}\rangle\,\langle \boldsymbol{y}, \boldsymbol{y}\rangle\,.$$

[5 marks]

Answers

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(c) Let **M** be a real symmetric matrix. Show that its eigenvalues are real, and that eigenvectors belonging to distinct eigenvalues are orthogonal.

Let $\mathbf{M}\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\boldsymbol{v} \neq 0$, allowing λ and $\boldsymbol{v}$ to be complex, and write $\boldsymbol{v}^*$ for the conjugate transpose. Then $\boldsymbol{v}^*\mathbf{M}\boldsymbol{v} = \lambda\,\boldsymbol{v}^*\boldsymbol{v}$. Taking the conjugate transpose of the whole expression and using $\mathbf{M}^* = \mathbf{M}$ (real and symmetric) gives $\boldsymbol{v}^*\mathbf{M}\boldsymbol{v} = \bar{\lambda}\,\boldsymbol{v}^*\boldsymbol{v}$. Since $\boldsymbol{v}^*\boldsymbol{v} = \|\boldsymbol{v}\|^2 > 0$ we may divide, giving $\lambda = \bar{\lambda}$, so λ is real. For the second part let $\mathbf{M}\boldsymbol{v}_1 = \lambda_1\boldsymbol{v}_1$ and $\mathbf{M}\boldsymbol{v}_2 = \lambda_2\boldsymbol{v}_2$ with $\lambda_1 \neq \lambda_2$. Then

$$\boldsymbol{v}_2^{\mathsf{T}}\mathbf{M}\boldsymbol{v}_1 = \lambda_1\,\boldsymbol{v}_2^{\mathsf{T}}\boldsymbol{v}_1, \qquad \boldsymbol{v}_2^{\mathsf{T}}\mathbf{M}\boldsymbol{v}_1 = (\mathbf{M}\boldsymbol{v}_2)^{\mathsf{T}}\,\boldsymbol{v}_1 = \lambda_2\,\boldsymbol{v}_2^{\mathsf{T}}\boldsymbol{v}_1,$$

using symmetry in the second. Subtracting, $(\lambda_1 - \lambda_2)\;\boldsymbol{v}_2^{\mathsf{T}}\boldsymbol{v}_1 = 0$, and as $\lambda_1 \neq \lambda_2$ we must have $\boldsymbol{v}_2^{\mathsf{T}}\boldsymbol{v}_1 = 0$.

[5 marks]

(d) Using only the triangle inequality $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ and homogeneity $\|a\,\boldsymbol{x}\| = |a|\|\boldsymbol{x}\|$, show that any proper norm $\|\cdot\|$ is a convex function.

Convexity requires $\|a\boldsymbol{x} + (1-a)\boldsymbol{y}\| \leq a\|\boldsymbol{x}\| + (1-a)\|\boldsymbol{y}\|$ for all $a \in [0,1]$. The triangle inequality gives

$$\|a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y}\| \leq \|a\,\boldsymbol{x}\| + \|(1-a)\,\boldsymbol{y}\|,$$

and by homogeneity, since $a \geq 0$ and $1 - a \geq 0$, this equals $a\|\boldsymbol{x}\| + (1-a)\|\boldsymbol{y}\|$, which is the required inequality.

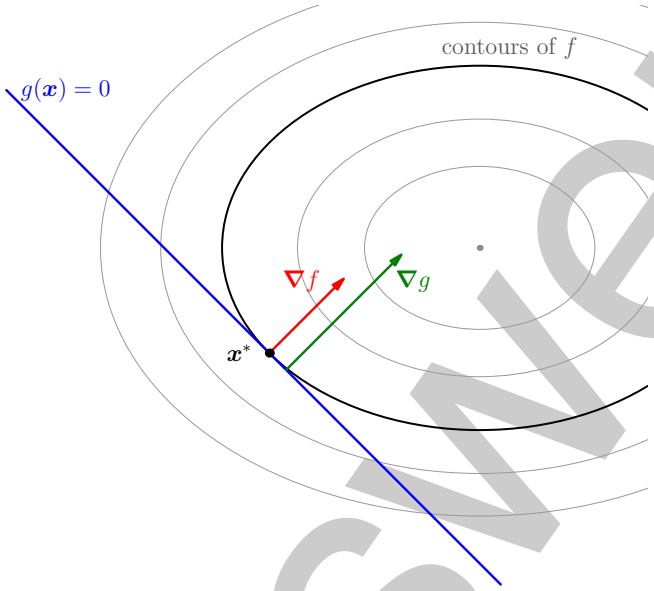
[5 marks]

Answers

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(e) We wish to maximise $f(\boldsymbol{x})$ subject to the constraint $g(\boldsymbol{x}) = 0$. Explain why at the constrained optimum the condition $\boldsymbol{\nabla} f = \lambda \boldsymbol{\nabla} g$ must hold for some scalar λ. Illustrate your answer with a sketch showing contours of f and the constraint curve.

Let δ be any small displacement that stays on the constraint surface, so that to first order $\delta^{\mathsf{T}} \nabla g = 0$; that is, the feasible displacements are exactly those orthogonal to ∇g. If x^* is a constrained maximum then f cannot increase to first order along any such direction, so $\delta^{\mathsf{T}} \nabla f = 0$ for every δ orthogonal to ∇g. A vector orthogonal to every vector orthogonal to ∇g must be parallel to ∇g, hence $\nabla f = \lambda \nabla g$. Geometrically the contour of f through x^* is tangent to the constraint curve: were they to cross, we could move along the constraint onto a higher contour.



[5 marks]

Answers

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(f) By considering the square of the integral and changing to polar coordinates, show that for $a > 0$

$$\int_{-\infty}^{\infty} \mathrm{e}^{-a\,x^2}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}.$$

Hence write down the normalisation constant of a normal distribution with variance σ^2.

Let I denote the integral. Writing the square as a two-dimensional integral and changing to polar coordinates, in which $x^2 + y^2 = r^2$ and $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$,

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-a\left(x^2+y^2\right)}\mathrm{d}x\mathrm{d}y = \int_0^{2\pi}\int_0^{\infty} \mathrm{e}^{-ar^2}\,r\,\mathrm{d}r\,\mathrm{d}\theta.$$

The angular integral gives 2π, and substituting $u = r^2$ (so $\mathrm{d}u = 2r\,\mathrm{d}r$) the radial integral is $\int_0^{\infty} \mathrm{e}^{-au}\mathrm{d}u/2 = 1/(2a)$. Hence $I^2 = 2\pi/(2a) = \pi/a$ and $I = \sqrt{\pi/a}$. A normal distribution has $\mathrm{e}^{-x^2/(2\sigma^2)}$, i.e. $a = 1/(2\sigma^2)$, so the integral is $\sqrt{2\pi\sigma^2}$ and the normalisation constant is $1/\sqrt{2\pi\sigma^2}$.

[5 marks]

(g) We observe n successes in N independent trials, each succeeding with unknown probability p, giving the binomial likelihood $\mathbb{P}(n|p) \propto p^n\,(1-p)^{N-n}$. Show that the beta distribution

$$\mathrm{Beta}(p|a,b) = \frac{p^{a-1}\,(1-p)^{b-1}}{B(a,b)}$$

is a conjugate prior to this likelihood, and state the parameters of the posterior.

By Bayes' rule the posterior is proportional to the likelihood times the prior,

$$\mathbb{P}(p|n) \propto p^n\,(1-p)^{N-n} \times p^{a-1}\,(1-p)^{b-1} = p^{a+n-1}\,(1-p)^{b+N-n-1}\,.$$

This is again of beta form, so the posterior is $\mathrm{Beta}\,(p\,|\,a+n,\ b+N-n)$, with normalisation $1/B(a+n, b+N-n)$ since it must be a normalised density. Prior and posterior lie in the same family, which is what is meant by conjugacy: the successes are added to a and the failures to b.

[5 marks]

Answers

TURN OVER

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(h) The Kullback–Leibler divergence between two distributions is

$$\mathrm{KL}\,(p\,\|\,q) = \sum_x p(x)\log\left(\frac{p(x)}{q(x)}\right).$$

Using Jensen's inequality, $\mathbb{E}\left[\log(Z)\right] \le \log(\mathbb{E}\left[Z\right])$, show that $\mathrm{KL}\,(p\,\|\,q) \ge 0$, and state when equality holds.

Write the divergence with the ratio inverted,

$$\mathrm{KL}\,(p\,\|\,q) = -\sum_x p(x)\log\left(\frac{q(x)}{p(x)}\right) = -\mathbb{E}\left[\log(Z)\right], \qquad Z = \frac{q(x)}{p(x)},$$

where the expectation is over x drawn from p. Jensen's inequality for the concave logarithm gives

$$\mathbb{E}\left[\log(Z)\right] \le \log\big(\mathbb{E}\left[Z\right]\big) = \log\left(\sum_x p(x)\frac{q(x)}{p(x)}\right) = \log\left(\sum_x q(x)\right) = \log(1) = 0.$$

Hence $\mathrm{KL}\,(p\,\|\,q) = -\mathbb{E}\left[\log(Z)\right] \ge 0$. Equality in Jensen's inequality for a strictly concave function requires Z to be constant, and since $\sum_x q(x) = \sum_x p(x) = 1$ that constant must be 1, so equality holds exactly when $p = q$.

[5 marks]

Answers

End of question A1

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Section B

B2 We have m data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m \in \mathbb{R}^p$ with mean $\boldsymbol{\mu}$ and covariance matrix

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^{m} (\boldsymbol{x}_k - \boldsymbol{\mu}) (\boldsymbol{x}_k - \boldsymbol{\mu})^{\mathsf{T}}.$$

(a) The variance of the data projected onto a unit vector $\boldsymbol{v}$ is $\sigma^2(\boldsymbol{v}) = \frac{1}{m-1} \sum_{k=1}^{m} \left(\boldsymbol{v}^{\mathsf{T}} (\boldsymbol{x}_k - \boldsymbol{\mu})\right)^2$. Using a Lagrange multiplier λ for the constraint $\|\boldsymbol{v}\|^2 = 1$, show that the direction of maximum variance satisfies $\mathbf{C}\boldsymbol{v} = \lambda \boldsymbol{v}$. Show that λ is the variance in that direction, and hence say which eigenvector of $\mathbf{C}$ is the first principal component. Explain why every eigenvalue of $\mathbf{C}$ is non-negative.

Since $\left(\boldsymbol{v}^{\mathsf{T}} \boldsymbol{z}\right)^2 = \boldsymbol{v}^{\mathsf{T}} \boldsymbol{z}\, \boldsymbol{z}^{\mathsf{T}} \boldsymbol{v}$,

$$\sigma^2(\boldsymbol{v}) = \boldsymbol{v}^{\mathsf{T}} \left(\frac{1}{m-1} \sum_{k=1}^{m} (\boldsymbol{x}_k - \boldsymbol{\mu}) (\boldsymbol{x}_k - \boldsymbol{\mu})^{\mathsf{T}} \right) \boldsymbol{v} = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v}.$$

The Lagrangian is $\mathcal{L} = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} - \lambda \left(\boldsymbol{v}^{\mathsf{T}} \boldsymbol{v} - 1\right)$, and as $\mathbf{C}$ is symmetric

$$\nabla_{\boldsymbol{v}} \mathcal{L} = 2\mathbf{C}\boldsymbol{v} - 2\lambda \boldsymbol{v} = \mathbf{0} \qquad \Longrightarrow \qquad \mathbf{C}\boldsymbol{v} = \lambda \boldsymbol{v},$$

so the stationary directions are the eigenvectors of $\mathbf{C}$. For such a direction $\sigma^2(\boldsymbol{v}) = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} = \lambda\, \boldsymbol{v}^{\mathsf{T}} \boldsymbol{v} = \lambda$: the multiplier is the variance along $\boldsymbol{v}$. The variance is therefore largest for the eigenvector with the largest eigenvalue, which is the first principal component.
For any $\boldsymbol{v}$, $\boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} = \sigma^2(\boldsymbol{v})$ is a sum of squares divided by $m-1$, so it is ≥ 0: $\mathbf{C}$ is positive semi-definite. Taking $\boldsymbol{v}$ to be a unit eigenvector gives $\lambda = \boldsymbol{v}^{\mathsf{T}} \mathbf{C} \boldsymbol{v} \geq 0$.

[10 marks]

Answers

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(b) Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ be orthonormal eigenvectors of **C** with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. Each data point is approximated by its projection onto the first K principal components,

$$\hat{\boldsymbol{x}}_k = \boldsymbol{\mu} + \sum_{i=1}^{K} \boldsymbol{v}_i \left(\boldsymbol{v}_i^{\mathsf{T}} \left(\boldsymbol{x}_k - \boldsymbol{\mu} \right) \right).$$

Show that the mean squared reconstruction error is

$$\frac{1}{m-1} \sum_{k=1}^{m} \| \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k \|^2 = \sum_{i=K+1}^{p} \lambda_i.$$

The eigenvectors form an orthonormal basis, so $\boldsymbol{x}_k - \boldsymbol{\mu} = \sum_{i=1}^{p} \boldsymbol{v}_i \boldsymbol{v}_i^{\mathsf{T}} (\boldsymbol{x}_k - \boldsymbol{\mu})$ and the residual is

$$\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k = \sum_{i=K+1}^{p} \boldsymbol{v}_i \, a_{ki}, \qquad a_{ki} = \boldsymbol{v}_i^{\mathsf{T}} \left(\boldsymbol{x}_k - \boldsymbol{\mu} \right).$$

By orthonormality $\| \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k \|^2 = \sum_{i=K+1}^{p} a_{ki}^2$. Averaging over the data,

$$\frac{1}{m-1} \sum_{k=1}^{m} \| \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k \|^2 = \sum_{i=K+1}^{p} \frac{1}{m-1} \sum_{k=1}^{m} a_{ki}^2 = \sum_{i=K+1}^{p} \boldsymbol{v}_i^{\mathsf{T}} \mathbf{C} \boldsymbol{v}_i = \sum_{i=K+1}^{p} \lambda_i,$$

using part (a) for the variance along each $\boldsymbol{v}_i$. Keeping the directions of largest variance therefore also minimises the reconstruction error.

[5 marks]

(c) Let **X** be the $p \times m$ matrix whose columns are the centred data $\boldsymbol{x}_k - \boldsymbol{\mu}$, so that $\mathbf{C} = \mathbf{X}\mathbf{X}^{\mathsf{T}}/(m-1)$, and let $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^{\mathsf{T}}$ be its singular value decomposition, with **U** and **V** orthogonal and **S** diagonal with singular values s_i. Show that the columns of **U** are eigenvectors of **C** and find the corresponding eigenvalues. Give one reason for computing the principal components from the SVD of **X** rather than by forming **C**.

Using $\mathbf{V}^{\mathsf{T}}\mathbf{V} = \mathbf{I}$,

$$\mathbf{C} = \frac{1}{m-1} \mathbf{U}\mathbf{S}\mathbf{V}^{\mathsf{T}}\mathbf{V}\mathbf{S}^{\mathsf{T}}\mathbf{U}^{\mathsf{T}} = \mathbf{U} \, \frac{\mathbf{S}\mathbf{S}^{\mathsf{T}}}{m-1} \, \mathbf{U}^{\mathsf{T}}.$$

$\mathbf{S}\mathbf{S}^{\mathsf{T}}$ is diagonal with entries s_i^2 (and zeros beyond the rank), so multiplying on the right by U gives $\mathbf{C}\mathbf{U} = \mathbf{U}\,\mathbf{S}\mathbf{S}^{\mathsf{T}}/(m-1)$. Column by column, $\mathbf{C}\boldsymbol{u}_i = \frac{s_i^2}{m-1}\boldsymbol{u}_i$: the columns of U are the principal components and the variances are $\lambda_i = s_i^2/(m-1)$.

Reasons to use the SVD (any one): forming $\mathbf{X}\mathbf{X}^{\mathsf{T}}$ squares the condition number, so small variances are lost to rounding error; when $p \gg m$ (images, genomes) C is a huge $p \times p$ matrix, while the economical SVD of X needs only $O(p\,m^2)$ work.

[8 marks]

Answers

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(d) Two-dimensional data have covariance matrix

$$\mathbf{C} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

Find the principal components and the variance along each. What fraction of the total variance does the first principal component explain, and what is the mean squared reconstruction error when the data are projected onto it?

The characteristic equation is $(3-\lambda)^2 - 1 = 0$, so $\lambda = 3 \pm 1$, giving $\lambda_1 = 4$ and $\lambda_2 = 2$. For $\lambda_1 = 4$, $(\mathbf{C} - 4\mathbf{I})\boldsymbol{v} = \mathbf{0}$ gives $v_1 = v_2$, so $\boldsymbol{v}_1 = (1,1)^\mathsf{T}/\sqrt{2}$; for $\lambda_2 = 2$, $v_1 = -v_2$, so $\boldsymbol{v}_2 = (1,-1)^\mathsf{T}/\sqrt{2}$. (They are orthogonal, as they must be for a symmetric matrix.)
The total variance is $\operatorname{tr}\,\mathbf{C} = 6 = \lambda_1 + \lambda_2$, so the first component explains $4/6 = 2/3$ of it. By part (b) the mean squared reconstruction error with $K = 1$ is the discarded eigenvalue, $\lambda_2 = 2$.

[7 marks]

Answers

End of question B2

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B3 A learning machine is trained on a data set $\mathcal{D}$ of m examples drawn independently from a distribution $\mu(\boldsymbol{x}, y)$, giving a predictor $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})$ with generalisation loss

$$L_G(\boldsymbol{\theta}_\mathcal{D}) = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y) \left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - y\right)^2 .$$

The *mean machine* is $\hat{f}_m(\boldsymbol{x}) = \mathbb{E}_\mathcal{D}\left[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})\right]$.

(a) Show that the expected generalisation loss $\bar{L}_G = \mathbb{E}_\mathcal{D}[L_G(\boldsymbol{\theta}_\mathcal{D})]$ can be written as $\bar{L}_G = B + V$, where

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y) \left(\hat{f}_m(\boldsymbol{x}) - y\right)^2 \qquad V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y)\, \mathbb{E}_\mathcal{D}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\right)^2\right].$$

Explain in words what each term measures, and how each typically changes as the model is made more complex.

The sum over (x, y) does not depend on $\mathcal{D}$, so the expectation can be taken inside. Adding and subtracting the mean machine,

$$\mathbb{E}_\mathcal{D}\left[\left(\hat{f} - y\right)^2\right] = \mathbb{E}_\mathcal{D}\left[\left(\hat{f} - \hat{f}_m\right)^2\right] + \left(\hat{f}_m - y\right)^2 + 2\,\mathbb{E}_\mathcal{D}\left[\hat{f} - \hat{f}_m\right]\left(\hat{f}_m - y\right),$$

where $\hat{f} = \hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})$ and we used that $\hat{f}_m(\boldsymbol{x}) - y$ does not depend on $\mathcal{D}$. The cross term vanishes because $\mathbb{E}_\mathcal{D}\left[\hat{f} - \hat{f}_m\right] = \hat{f}_m - \hat{f}_m = 0$. Weighting by $\mu(\boldsymbol{x}, y)$ and summing gives $\bar{L}_G = B + V$.
B, the bias, is the loss of the mean machine: how well the model class can fit even when the luck of the particular data set is averaged away (it also includes the irreducible noise in y). V, the variance, measures how much the learned predictor changes from one data set to another. A more complex model usually has smaller bias, since it can represent the target better, but larger variance, since it can also fit the noise of each data set: the bias–variance dilemma. More data reduces V.

[10 marks]

Answers

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(b) Consider the simplest learning problem: estimating the mean θ of a distribution with variance σ^2 from m independent samples with sample mean $\bar{x}$, using the *shrinkage* estimator $\hat{\theta} = c\,\bar{x}$ with $0 \le c \le 1$. (Shrinkage is used in statistics to reduce the sample estimate error in systems with high variance. It deliberately underestimates the mean.) Find $\mathbb{E}\left[\hat{\theta}\right]$ and $\mathrm{Var}(\hat{\theta})$, and hence show that

$$\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] = (1-c)^2\,\theta^2 + c^2\,\frac{\sigma^2}{m}.$$

$\mathbb{E}\left[\bar{x}\right] = \theta$ and, the samples being independent, $\mathrm{Var}(\bar{x}) = \sigma^2/m$. Hence $\mathbb{E}\left[\hat{\theta}\right] = c\,\theta$ and $\mathrm{Var}(\hat{\theta}) = c^2\sigma^2/m$. As in part (a), adding and subtracting $\mathbb{E}\left[\hat{\theta}\right]$ and noting that the cross term has zero mean,

$$\mathbb{E}\left[\left(\hat{\theta}-\theta\right)^2\right] = \left(\mathbb{E}\left[\hat{\theta}\right]-\theta\right)^2 + \mathrm{Var}(\hat{\theta}) = (1-c)^2\theta^2 + c^2\frac{\sigma^2}{m}:$$

a squared bias plus a variance.

[5 marks]

(c) Find the value c^* that minimises the mean squared error in part (b). Show that $0 < c^* < 1$ whenever $\theta \neq 0$, and that the minimum mean squared error is smaller than σ^2/m, the error of the unbiased estimate $\bar{x}$. Why can c^* not be used directly in practice, and what happens to it as $m \to \infty$?

Differentiating,

$$\frac{\mathrm{d}}{\mathrm{d}c}\left((1-c)^2\theta^2 + c^2\frac{\sigma^2}{m}\right) = -2(1-c)\,\theta^2 + 2c\,\frac{\sigma^2}{m} = 0 \quad\Longrightarrow\quad c^* = \frac{\theta^2}{\theta^2+\sigma^2/m},$$

a minimum as the second derivative $2\theta^2 + 2\sigma^2/m$ is positive. Numerator and denominator are positive and the denominator is larger, so $0 < c^* < 1$. Writing $a = \theta^2$ and $b = \sigma^2/m$, the minimum is

$$\frac{b^2a + a^2b}{(a+b)^2} = \frac{ab}{a+b} = c^*\,\frac{\sigma^2}{m} < \frac{\sigma^2}{m}.$$

Accepting a little bias buys a larger reduction in variance. But c^* depends on the unknown θ, so it has to be estimated, or c treated as a hyper-parameter and chosen by cross-validation. As $m \to \infty$, $c^* \to 1$: with plenty of data the variance is small and shrinkage is no longer worth its bias.

[8 marks]

Answers

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(d) Ridge regression in one dimension: targets $y_k = w\,x_k + \epsilon_k$ are observed at fixed inputs x_k, $k = 1, \ldots, m$, with independent noise of mean zero and variance σ^2. The estimate $\hat{w}$ minimises $\sum_k (y_k - w\,x_k)^2 + \eta\,w^2$. Writing $S = \sum_k x_k^2$, show that $\hat{w} = \sum_k x_k y_k/(S+\eta)$, find its mean and variance, and explain how η trades bias against variance.

Setting the derivative to zero, $-2\sum_k x_k\,(y_k - \hat{w}x_k) + 2\eta\hat{w} = 0$, so $\hat{w}\,(S+\eta) = \sum_k x_k y_k$. Substituting the model, $\sum_k x_k y_k = wS + \sum_k x_k \epsilon_k$, so

$$\mathbb{E}\left[\hat{w}\right] = \frac{S}{S+\eta}\,w \qquad\qquad \mathrm{Var}(\hat{w}) = \frac{\sigma^2 S}{(S+\eta)^2}.$$

At $\eta = 0$ the estimate is unbiased with variance σ^2/S. Increasing η shrinks $\hat{w}$ towards zero, introducing a bias $-\eta w/(S+\eta)$ but reducing the variance. This is the shrinkage estimator of part (b) with $c = S/(S+\eta)$, so as there a moderate η lowers the expected error; it is chosen by cross-validation.

[7 marks]

Answers

End of question B3

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B4 In Bayesian inference we want expectations under the posterior $\pi(\boldsymbol{\theta}) = \mathbb{P}(\mathcal{D}|\boldsymbol{\theta})\,\mathbb{P}(\boldsymbol{\theta})\,/\mathbb{P}(\mathcal{D})$ of a parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$.

(a) Explain why computing a posterior expectation by numerical integration on a grid becomes infeasible as d grows. Show that the Monte Carlo estimate $\hat{g} = \frac{1}{n}\sum_{i=1}^{n} g(\boldsymbol{\theta}_i)$, from n independent samples drawn from π, is unbiased and has a standard error proportional to $1/\sqrt{n}$ whatever the value of d.

A grid with k points along each axis has k^d points, which grows exponentially with d; even the evidence $\mathbb{P}(\mathcal{D})$ is such an integral. For the Monte Carlo estimate, $\mathbb{E}\left[\hat{g}\right] = \frac{1}{n}\sum_i \mathbb{E}\left[g(\boldsymbol{\theta}_i)\right] = \mathbb{E}\left[g\right]$, so it is unbiased, and by independence

$$\mathrm{Var}(\hat{g}) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}\left(g(\boldsymbol{\theta}_i)\right) = \frac{\mathrm{Var}(g)}{n},$$

a standard error $\sqrt{\mathrm{Var}(g)}/\sqrt{n}$. The dimension enters only through $\mathrm{Var}(g)$, not through the rate.

[5 marks]

(b) A Markov chain with transition density $W(\boldsymbol{\theta} \to \boldsymbol{\theta}')$ satisfies *detailed balance* with respect to π if $W(\boldsymbol{\theta} \to \boldsymbol{\theta}')\,\pi(\boldsymbol{\theta}) = W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\pi(\boldsymbol{\theta}')$.

(i) Show that π is then a stationary distribution of the chain.

(ii) The Metropolis–Hastings algorithm proposes $\boldsymbol{\theta}'$ from a proposal density $q(\boldsymbol{\theta}'|\boldsymbol{\theta})$ and accepts it with probability

$$A(\boldsymbol{\theta} \to \boldsymbol{\theta}') = \min\left(1, \frac{\pi(\boldsymbol{\theta}')\,q(\boldsymbol{\theta}|\boldsymbol{\theta}')}{\pi(\boldsymbol{\theta})\,q(\boldsymbol{\theta}'|\boldsymbol{\theta})}\right).$$

Show that the resulting chain satisfies detailed balance, and explain why the evidence $\mathbb{P}(\mathcal{D})$ is not needed.

(i) Integrating detailed balance over $\boldsymbol{\theta}$, and using that the chain must go somewhere, $\int W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = 1$,

$$\int W(\boldsymbol{\theta} \to \boldsymbol{\theta}')\,\pi(\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = \pi(\boldsymbol{\theta}') \int W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = \pi(\boldsymbol{\theta}').$$

The left side is the distribution after one step starting from π, so π is unchanged by the chain.
(ii) For $\boldsymbol{\theta}' \neq \boldsymbol{\theta}$, $W(\boldsymbol{\theta} \to \boldsymbol{\theta}') = q(\boldsymbol{\theta}'|\boldsymbol{\theta})\,A(\boldsymbol{\theta} \to \boldsymbol{\theta}')$. Let r be the ratio inside the minimum and suppose $r \leq 1$; then $A(\boldsymbol{\theta} \to \boldsymbol{\theta}') = r$ while the reverse ratio is $1/r \geq 1$, so $A(\boldsymbol{\theta}' \to \boldsymbol{\theta}) = 1$. Hence

$$W(\boldsymbol{\theta} \to \boldsymbol{\theta}')\,\pi(\boldsymbol{\theta}) = q(\boldsymbol{\theta}'|\boldsymbol{\theta})\,\frac{\pi(\boldsymbol{\theta}')\,q(\boldsymbol{\theta}|\boldsymbol{\theta}')}{\pi(\boldsymbol{\theta})\,q(\boldsymbol{\theta}'|\boldsymbol{\theta})}\,\pi(\boldsymbol{\theta}) = q(\boldsymbol{\theta}|\boldsymbol{\theta}')\,\pi(\boldsymbol{\theta}') = W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\pi(\boldsymbol{\theta}').$$

The case $r > 1$ is the same with the roles exchanged, and a rejected move ($\boldsymbol{\theta}' = \boldsymbol{\theta}$) satisfies detailed balance trivially. The acceptance depends on π only through the ratio $\pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta})$, in which $\mathbb{P}(\mathcal{D})$ cancels: we need only likelihood times prior.

[10 marks]

TURN OVER

Answers

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(c) A Markov chain has three states with target probabilities $\boldsymbol{\pi} = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)^{\mathsf{T}}$. From state j it proposes one of the other two states, each with probability $\frac{1}{2}$, and accepts a proposed state i with probability $\min(1, \pi_i/\pi_j)$, otherwise staying at j. Write down the transition matrix $\mathbf{M}$, where M_{ij} is the probability of moving from state j to state i, and verify that $\mathbf{M}\boldsymbol{\pi} = \boldsymbol{\pi}$.

From state 1: to 2 with $\frac{1}{2} \times \frac{1/3}{1/2} = \frac{1}{3}$, to 3 with $\frac{1}{2} \times \frac{1/6}{1/2} = \frac{1}{6}$, so it stays with $\frac{1}{2}$.
From state 2: to 1 with $\frac{1}{2}$ (always accepted), to 3 with $\frac{1}{2} \times \frac{1/6}{1/3} = \frac{1}{4}$, stays with $\frac{1}{4}$.
From state 3: both moves are uphill, so it moves to 1 or 2 with $\frac{1}{2}$ each and never stays. Hence

$$\mathbf{M} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{6} & \frac{1}{4} & 0 \end{pmatrix},$$

whose columns each sum to 1. Then

$$\mathbf{M}\boldsymbol{\pi} = \begin{pmatrix} \frac{1}{4} + \frac{1}{6} + \frac{1}{12} \\ \frac{1}{6} + \frac{1}{12} + \frac{1}{12} \\ \frac{1}{12} + \frac{1}{12} + 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{6} \end{pmatrix} = \boldsymbol{\pi}.$$

[8 marks]

(d) Show that for a symmetric proposal, $q(\boldsymbol{\theta}'|\boldsymbol{\theta}) = q(\boldsymbol{\theta}|\boldsymbol{\theta}')$, the acceptance probability becomes $\min(1, \pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta}))$. For a target $\mathcal{N}(0,1)$ and a random-walk proposal $\theta' = \theta + s\,\epsilon$ with $\epsilon \sim \mathcal{N}(0,1)$, find the probability of accepting a proposed move from $\theta = 0$ to $\theta' = 2$, and from $\theta = 2$ to $\theta' = 0$. Explain how the step size s affects the efficiency of the sampler, and what is meant by burn-in.

With q symmetric the proposal densities cancel from the ratio, leaving $\min(1, \pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta}))$; the random walk is symmetric because ϵ and $-\epsilon$ are equally likely. For $\mathcal{N}(0,1)$, $\pi(2)/\pi(0) = \mathrm{e}^{-2} \approx 0.135$, so the move away from the mode is accepted with probability 0.135, while the move from 2 to 0 has ratio $\mathrm{e}^{2} > 1$ and is always accepted.
If s is very small almost every move is accepted but each moves only a little, so the chain diffuses slowly and successive samples are highly correlated. If s is very large most proposals land where π is tiny and are rejected, so the chain sits still. Efficiency is best in between, at a moderate acceptance rate (roughly a quarter to a half). Burn-in is the initial stretch of the chain, before it has forgotten its starting point and reached the stationary distribution; these samples are discarded.

[7 marks]

End of question B4

Answers

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END OF PAPER

This page contains no examination material

Answers